

Homework 4

STATISTICAL MODELING AND LEARNING

due: **Friday, February 27, 2026 at 10:00a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

Problem 1: Model Selection and Overfitting. Download the file `greenbuildings.csv` from the course website. This dataset contains commercial real estate listings. The outcome is `Rent` (rent per square foot). Use the following numeric predictors: `size`, `empl_gr`, `leasing_rate`, `stories`, `age`, `renovated`, `class_a`, `class_b`, `green_rating`, `net`, `amenities`, `cd_total_07`, `hd_total07`, `Precipitation`, `Gas_Costs`, `Electricity_Costs`. (Exclude `CS_PropertyID`, `cluster`, `LEED`, `Energystar`, `total_dd_07`, and `cluster_rent`.)

- The AIC is defined as $AIC = 2k - 2\ln(L)$, where k is the number of parameters and L is the likelihood. For linear regression with normal errors, this simplifies to $AIC = 2k + n\ln(RSS/n)$. Explain in one or two sentences why AIC tends to decrease when we add a predictor, but the penalty term $2k$ prevents us from always preferring the largest model.
- Fit the full model (all 17 predictors) and use backward stepwise regression to select a model. Report the final model chosen by stepwise selection. How many predictors does it retain?
- Split the data into a training set (70%) and test set (30%), using `set.seed(42)` for reproducibility. Fit both the full model and the reduced model from part (b) on the training set. Compute the RMSE on the training set and on the test set for each model. Create a table comparing the two models. Which model generalizes better to the test set? What does this tell you about overfitting?
- Use best subsets regression (`regsubsets` from the `leaps` package) to find the best model for each number of predictors (1 through 17). Plot Adjusted R^2 versus number of predictors. How does the best subset at the stepwise-chosen size compare to the stepwise result from part (b)?

Problem 2: Penalized Regression. Download the file `social_marketing.csv` from the course website. This dataset contains Twitter engagement data: each row is a user, and columns indicate the proportion of their tweets in various categories (e.g., `chatter`, `travel`, `politics`, `food`). The outcome is `spam` (0, 1, or 2). Create a binary outcome: $y = 1$ if `spam` > 0, else $y = 0$. We will predict y from the other 35 engagement categories using logistic regression with penalization.

- In penalized logistic regression, we minimize $-\ell(\beta) + \lambda P(\beta)$, where ℓ is the log-likelihood and $P(\beta)$ is the penalty. Ridge uses $P(\beta) = \sum_j \beta_j^2$; LASSO uses $P(\beta) = \sum_j |\beta_j|$. In one or two sentences, explain why LASSO can produce coefficients that are exactly zero (variable selection), while Ridge typically does not.
- Prepare the data: create a model matrix X of the 35 predictor columns (exclude the first column, which is user ID, and `spam`). Create the binary response y as above. Remove any rows with missing values. Center and scale the predictors. Split the data 70/30 into training and test sets with `set.seed(42)`.

- c. Fit Ridge-penalized logistic regression using `glmnet` with `alpha = 0` and `family = "binomial"`. Use 10-fold cross-validation to select λ . Report $\lambda_{\min}$. Plot the coefficient paths. Do any coefficients go exactly to zero?
- d. Fit LASSO-penalized logistic regression using `glmnet` with `alpha = 1` and `family = "binomial"`. Use 10-fold CV. Report $\lambda_{\min}$ and λ_{1se} . How many predictors have nonzero coefficients at $\lambda_{\min}$? At λ_{1se} ?
- e. Compare unpenalized logistic regression, Ridge (at $\lambda_{\min}$), and LASSO (at $\lambda_{\min}$) on the test set. Report test set accuracy (proportion correctly classified) and, if applicable, AUC. Which method performs best? Which engagement categories does LASSO identify as most predictive of spam?

Problem 3: Regression Trees and Ensembles. Download the file `wine.csv` from the course website. This dataset contains chemical properties of red and white wines. The outcome is `quality` (an integer score from 0–10). Predictors are the numeric chemical variables: `fixed.acidity`, `volatile.acidity`, `citric.acid`, `residual.sugar`, `chlorides`, `free.sulfur.dioxide`, `total.sulfur.dioxide`, `density`, `pH`, `sulphates`, `alcohol`.

- a. In a regression tree, each leaf predicts the mean of Y for observations in that leaf. The tree is built by recursively splitting to minimize RSS. In one or two sentences, explain why a very deep tree might have low training error but high test error. How does this relate to the bias–variance tradeoff?
- b. Fit a regression tree using `rpart` with `method = "anova"`. Use the default settings. Plot the tree. How many leaves does it have? Interpret the top split: which chemical property and cutoff are most important for predicting `quality`?
- c. Use `printcp()` or `plotcp()` to examine cross-validated error. Prune the tree to minimize cross-validated error. How many leaves does the pruned tree have? Plot the pruned tree.
- d. Split the data 70/30 (use `set.seed(42)`). Fit the full tree and the pruned tree on the training set. Compute test set RMSE for both. Which performs better? Why might pruning help?
- e. Fit a bagged ensemble of regression trees using the `randomForest` package with `mtry = p` (use all predictors at each split, so each tree is grown on a bootstrap sample of the data). Use `ntree = 100` and `set.seed(42)`. Compute test set RMSE for the bagged model. Compare to the pruned single tree from part (d). Which performs better? In one or two sentences, explain why bagging can reduce variance compared to a single tree.

Problem 4: Neural Networks. Continue using the `wine.csv` dataset. We will predict `quality` (an integer score from 0–10) from the chemical predictors. Use the `nnet` package.

- a. Draw a diagram of a neural network with 2 inputs (x_1, x_2), 2 hidden units, and 1 output $\hat{y}$. Label the weights (e.g., w_1, w_2, v_{11}, v_{12} , etc.) and show how the output is a composition of functions from inputs to hidden layer to output.
- b. For the network in part (a), write $\hat{y}$ explicitly as a composition: $\hat{y} = g(\cdots f(\cdots x_1, x_2 \cdots) \cdots)$. Identify where the chain rule would be used when computing gradients for gradient descent.

- c. In one or two sentences, explain why a neural network with many hidden units can approximate complex nonlinear relationships, whereas a linear model cannot.
- d. Prepare the data: use the chemical predictors (`fixed.acidity` through `alcohol`) and outcome `quality`. Scale all predictors to mean 0 and standard deviation 1 (use the training set to compute scaling, then apply to test set). Split 70/30 with `set.seed(42)`.
- e. Fit a neural network with one hidden layer and 5 hidden units. Use `nnet(..., size = 5, linout = TRUE)` for regression. Set `maxit = 500` and `trace = FALSE`. Report the final residual sum of squares. (Note: neural networks are sensitive to random initialization; results may vary slightly.)
- f. Compute test set RMSE for your neural network. Compare to a linear regression model and the pruned regression tree (predicting `quality` from the same predictors) on the test set. Create a brief summary table. Which method performs best? In one or two sentences, discuss when you might prefer a neural network over a linear model or tree.